Anindita Maiti

Curriculum Vitae

Contact

Perimeter Institute, Waterloo, ON, Postal Code N2L 2Y5, Canada. Personal website
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Academic Positions

Perimeter Institute for Theoretical Physics – Waterloo, Canada — Sept 2023 - Present
Postdoctoral Fellow
Mentor: Roger Melko

Harvard John A. Paulson SEAS – Boston, USA — May 2023 - Aug 2023
Postdoctoral Fellow (Applied Math)
Supervisor: Cengiz Pehlevan

Education

Northeastern University – Boston, USA — 2017-2023
Ph.D. (Physics)
Advisor: James Halverson

The NSF AI Institute for Artificial Intelligence and Fundamental Interactions – Boston
Junior Investigator — 2020-2023

Indian Institute of Technology Bombay – Mumbai, India — 2012-2017
Integrated Bachelor and Master of Technology (Engineering Physics) with Honors (Physics)
Advisor: Urjit Yajnik

Research Interests

Theoretical physics + applied mathematics for Artificial Intelligence (AI).
Bridging theory-practice gap in AI interpretability and robustness.
AI for quantum field theory and quantum.

Preprints & Publications

J. Q. Toledo-Marin, **A. Maiti**, G. C. Fox, R. G. Melko, "*Exploring the Energy Landscape of RBMs: Reciprocal Space Insights into Bosons, Hierarchical Learning and Symmetry Breaking*", [arXiv:2503.21536].

J. N. Howard, M. S. Klinger, **A. Maiti**, A. G. Stapleton, "*Bayesian RG Flow in Neural Network Field Theories*", SciPost Physics Core, 8(1), 027, [arXiv:2405.17538].

Y. M. Lu, M. I. Letey, J. A. Zavatone-Veth, **A. Maiti**, C. Pehlevan, "*Asymptotic theory of in-context learning by linear attention*", [arXiv:2405.11751].

Y. M. Lu, M. I. Letey, J. A. Zavatone-Veth, **A. Maiti**, C. Pehlevan, "*In-Context Learning by Linear Attention: Exact Asymptotics and Experiments*", [NeurIPS 2024 Workshop on Mathematics of Modern Machine Learning].

J. N. Howard, R. Jefferson, **A. Maiti**, Z. Ringel, "*Wilsonian Renormalization of Neural Network Gaussian Processes*", Mach. Learn. Sci. Tech., [arXiv:2405.06008].

M. Demirtas, J. Halverson, **A. Maiti**, M. D. Schwartz, K. Stoner, "*Neural Network Field Theories: Non-Gaussianity, Actions, and Locality*", Mach. Learn. Sci. Tech. 5, 015002 (2024), [arXiv:2307.03223].

A. Maiti, K. Stoner, J. Halverson, "*Symmetry-via-Duality: Invariant Neural Network Densities from Parameter-Space Correlators*", MACHINE LEARNING: IN PURE MATHEMATICS AND THEORETICAL PHYSICS, 2023, **293-330**, [arXiv:2106.00694v1].

J. Halverson, C. Long, **A. Maiti**, B. Nelson, G. Salinas, "*Gravitational waves from dark Yang-Mills sectors*", JHEP **05** (2021), 154, [arXiv:2012.04071].

J. Halverson, **A. Maiti**, K. Stoner, "*Neural Networks and Quantum Field Theory*", Mach. Learn. Sci. Tech. **2** (2021) no. 3, 035002, [arXiv:2008.08601].

In-Progress

"*Fermions and Susy in Neural Network Field Theories*" - with James Halverson, Fabian Ruehle, Samuel Frank.

"*Neural Network Field Theory for Neural Scaling Laws*" - with Tianji Cai, Yushao Chen, Roger Melko.

Awards & Honors

UC Riverside Chancellor's Postdoctoral Fellowship: Jul 1, 2023 to Jul 1, 2024 (Declined).

Dean's Graduate Student Excellence Award in Research: Northeastern University College of Science (Spring 2021).

Lawrence Award for Graduate Academic Excellence: Northeastern University Dept. of Physics (Spring 2018).

Travel Grants: "Theoretical Physics for Machine Learning" Workshop by Aspen Center for Physics (Feb 2023); "New Frontiers in Machine Learning and Quantum" Workshop by Perimeter Institute (Nov 2022); The NSF IAIFI (Feb 2023); Northeastern University Dept. of Physics (Summer 2022); Northeastern University PhD Network (Summer 2022).

Indian Academy of Sciences Summer Research Fellowship: Indian Academy of Sciences, (Summer 2014).

Seminars, Talks, Colloquia

Invited Workshop, Conference Talks, Panel Discussions

Probing the Frontiers of Nuclear Physics with AI at the EIC, Stony Brook University CFNS — Mar 2025

No Stone Unturned 2025, University of Utah, USA — Mar 2025

Panel Discussion: The Intersection of AI, Physics, and Tech Innovation, Perimeter Institute in partnership with Volta, Halifax, Nova Scotia, Canada — Feb 2025

String Data 2024, Kyoto University, Japan — Dec 2024

Panel Discussion: She Talks Tech, The Accelerator Center, Kitchener, Canada — July 2024

Machine Learning and the Renormalization Group, ECT*, Italy — May 2024

Future Horizons: Bridging AI, Quantum and New Materials, IVADO, Perimeter Institute, Institut Courtois and Mila Quebec, Canada — May 2024

Fields, Strings, and Deep Learning, Aspen Center for Physics — Jan 2024

(Non-speaking) Program Invitation: Deep Learning from the Perspective of Physics and Neuroscience, Kavli Institute for Theoretical Physics, UCSB — Nov 2023

AI and Quantum Information for Particle Physics, KAIST and IBS Center for Theoretical Physics of the Universe — Nov 2023

Probing the Frontiers of Nuclear Physics with AI at the EIC, Stony Brook University CFNS — Sept 2023

Machine Learning for Lattice Field Theory and Beyond, ECT*, Italy — Jun 2023

New Frontiers in Machine Learning and Quantum, Perimeter Institute — Nov 2022

Short Talks: A Deep-Learning Era of Particle Theory, Mainz Institute for Theoretical Physics, Johannes Gutenberg University [slides] — June 2022

String Data 2021, University of Witwatersrand & University of Cape Town [slides] — Dec 2021

Invited Seminars

High-TheQ group seminar series, Ghent University, Belgium — Jul 2025

CHEP Seminar, Indian Institute of Science, Bangalore — Jan 2025

Strings Seminar, International Center for Theoretical Sciences, Bangalore — Jan 2025

High Energy Physics Seminar, Indian Institute of Technology Kanpur — Jan 2025

High Energy Physics Seminar, Indian Institute of Technology Kanpur — June 2024

Stanford Linear Accelerator Center (SLAC) AI Seminar, Stanford University — Feb 2024

ORIGINS Data Science Lab Seminar, TU Munich — Jul 2023

Center for Theoretical Physics Seminar, Seoul National University — Mar 2023

AIC Seminar, Université Paris-Saclay, CEA-LIST Jan 2023

UCI Physics Astro/Particle-ML Seminar Series, UC Irvine Oct 2022

UCSB Joint HEX-HET Seminar Series, UC Santa Barbara Oct 2022

HEP Seminar, UC Riverside Oct 2022

Majorana-Raychaudhuri Seminar Series, INFN & University Salerno, Italy & PAMU, Indian Statistical Institute, Kolkata, India Sept 2022

Computational Algebra Seminar Series, University of Nottingham, UK Sept 2022

Pehlevan Research Group Journal Club, Harvard University [slides] Aug 2022

QFT Research Seminar, Institute for Theoretical Physics - Münster (WWU) [slides] May 2021

Joint High Energy Theory and Machine Learning Seminar, Heidelberg University, LMU Munich and Northeastern University [slides] May 2021

Journal Club, The NSF AI Institute for A. I. and Fundamental Interactions [slides] Feb 2021

Seminar Series on String Phenomenology [slides] Oct 2020

Contributed Workshop, Conference Talks, Posters

Parallel Session: Summer Workshop 2024, The NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) Aug 2024

String Data 2023, Caltech Dec 2023

Parallel Session: Summer Workshop 2023, The NSF AI Institute for Artificial Intelligence and Fundamental Interactions (IAIFI) Aug 2023

Poster Session: Theoretical Physics for Machine Learning, Aspen Center for Physics Feb 2023

Poster session: Summer Workshop 2022, The NSF IAIFI Aug 2022

Parallel Session: String Phenomenology 2022, University of Liverpool [slides] Jul 2022

Contributed Seminars

Machine Learning Initiative Seminar, Perimeter Institute Feb 2024

IPPP Seminar, Institute for Particle Physics Phenomenology, Durham University Nov 2022

Oxford Dalitz Seminar in Fundamental Physics, U. Oxford Nov 2022

Theoretical Particle Physics & Cosmology Seminar, King's College London Oct 2022

Mathematics Seminar, City, University of London Oct 2022

Theoretical Physics Seminar, Uppsala University Oct 2022

Journal Club, The NSF AI Institute for A. I. and Fundamental Interactions Sept 2022

Organization

Workshop and Conferences

A Look into the Future through the Lens of AI Research, Perimeter Institute for Theoretical Physics, Canada Apr 2025

- *Member of the principal organizing committee*

Generative AI for High & Low Energy Physics, Kavli Institute for Theoretical Physics, UC Santa Barbara, USA Nov-Dec 2025

- *Member of the organizing committee and DEI coordinator*

Future Horizons: Bridging AI, Quantum and New Materials, IVADO, Perimeter Instittue, Institut Courtois and Mila Quebec, Canada May 2024

- *Member of the organizing committee*

At the interface of physics, mathematics and artificial intelligence, Pollica Physics Center, Italy May-June 2023

- *Member of the organizing committee*

Teaching Experience

TUTORIAL LEAD – Future Horizons: Bridging AI, Quantum and New Materials 2024
Machine Learning from a Physicist's Perspective: lectures by Anindita Maiti

TUTORIAL LEAD – The NSF IAIFI Summer School 2023
Normalizing Flows for Lattice Field Theory: lectures by Miranda Cheng

TEACHING ASSISTANT – Northeastern University, Boston, Massachusetts
PHYS 7325: Quantum Field theory 1 (Fall 2020, Fall 2019)
PHYS 5115: Quantum Mechanics (Spring 2020, Spring 2019)
PHYS 3601: Classical Dynamics (Fall 2018)
PHYS 2305: Thermo and Statistical Mechanics (Spring 2018)
PHYS 1155: Physics Lab for Engineering 2 (Fall 2017)
PHYS 3600: Advanced Physics Lab (multiple semesters)
Undergraduate Physics lab (multiple semesters)

TEACHING ASSISTANT – IIT Bombay, Mumbai, India
PH 117: Undergraduate Physics lab (Spring 2017)
EP 215: Undergraduate Electronics lab (Fall 2016)

Professional Service Activities

MEMBER, ANTI-RACISM WORKING GROUP & MENTAL HEALTH WORKING GROUP: Perimeter Institute for Theoretical Physics (Sept 2023 - Present).

ALUMNUS MENTOR: For undergraduate students in Engineering Physics major at IIT Bombay. (Sept 2022 - Dec 2024)

MEMBER, EARLY CAREER AND EQUITY COMMITTEE: The NSF AI Institute for Artificial Intelligence and Fundamental Interactions. (Jan 2021 - Dec 2022)

MEMBER, GRADUATE STUDENT COUNCIL: Northeastern University College of Science. (Sept 2020 - Aug 2022)

COORDINATOR & INITIATOR, GRADUATE WOMEN IN PHYSICS SOCIETY : Northeastern University Dept. of Physics. (Sept 2021 - May 2023)

REFEREE: SynS & ML @ ICML2023; NeurIPS 2022 workshop on Machine Learning and the Physical Sciences; NeurIPS 2021 workshop on Machine Learning and the Physical Sciences; 'Foundations of Physics' Journal; NeurIPS 2020 workshop on Machine Learning and the Physical Sciences.

Summer Schools

- IAIFI Summer School, Aug 2022, The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.
- Theoretical Advanced Study Institute in Particle Theory (TASI), June 2021, CU Boulder.
- Deep Learning Theory Summer School at Princeton, Jul 2021, Princeton University.

References

(1) PROF. ROGER MELKO, (*Email: rmelko@perimeterinstitute.ca*),
Professor, Department of Physics & Astronomy, University of Waterloo,
Associate Faculty, Perimeter Institute for Theoretical Physics,
Canada Research Chair in Computational Quantum Many-Body Physics

(2) PROF. JAMES HALVERSON, (*Email: j.halverson@northeastern.edu*),
Associate Professor, Dept. of Physics, Northeastern University,
The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

(3) PROF. CENGIZ PEHLEVAN, (*Email: cpehlevan@seas.harvard.edu*),
Assistant Professor, Dept. of Applied Mathematics, Harvard University.
Kempner Institute Associate Faculty.

(4) PROF. MATTHEW SCHWARTZ, (*Email: schwartz@g.harvard.edu*),
Associate Professor, Dept. of Physics, Harvard University,
The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

(5) Prof. Fabian Ruehle, (*Email: f.ruehle@northeastern.edu*),
Assistant Professor, Dept. of Physics and Dept. of Mathematics, Northeastern University,
The NSF AI Institute for Artificial Intelligence and Fundamental Interactions.

Technical skills

Programming languages: Python, C, C++, Mathematica, Matlab, Pytorch.